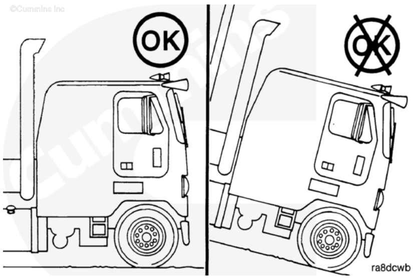
Trainings

Coolant Replacer Method

Evacuation

The following steps are used to evacuate the cooling system using the coolant replacer tool, Part Number 2892459.

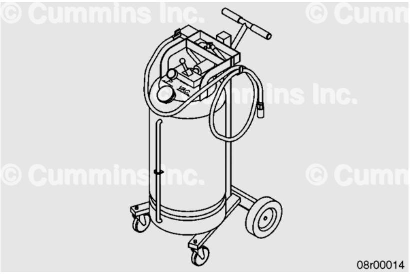


Note : When the vehicle/equipment or engine is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method is the preferred method for coolant removal. Use the coolant replacer tool, Part Number 2892459. If the vehicle/equipment or engine does not have a quick disconnect fitting presently installed, one can be installed in the cooling system package to utilize the Coolant Replacer Method; otherwise the coolant drain and fill method must be used.

Position the equipment on level ground.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Do not use the coolant replacement tool to evacuate contaminated coolant or a system that is suspected of contamination. This could result in cross contamination of coolant. The coolant drain method should be used for removing contaminated coolant. Refer to the manufacturer's manual for specific instructions on cleaning the tool of contaminates.

If the engine is found to have internal or external coolant leaks a coolant test **must** be performed and passed using an appropriate coolant test strip to use the coolant replacement tool. Use of coolant that is **not** acceptable for reuse can cause pitting and corrosion damage. Coolant that does not meet reuse guidelines **must** be replaced or brought into specification and tested by adjusting antifreeze, water, and supplemental coolant additive (SCA) levels if applicable. See Section 5 in the Fluids for Cummins® Products Service Manual, Bulletin 5411406, for antifreeze, water, and test strip specifications.

Note : See equipment manufacturer service information for special coolant drain requirements. Special instructions may also be located near the cooling system access point or fill door on the vehicle.

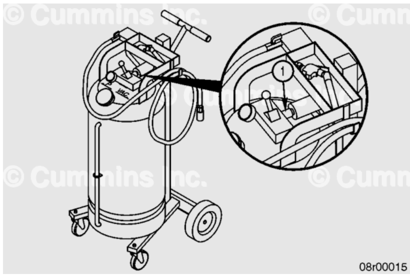
Isolate the engine from the vehicle cooling system by closing coolant flow valves to the equipment heating systems before starting the repair. This will prevent the heater circuit from draining, minimizing the chance for air pockets to be present during the fill process.

This air can be very difficult to purge in some applications with several feet of plumbing and multiple heater cores.

Remove the radiator cap.

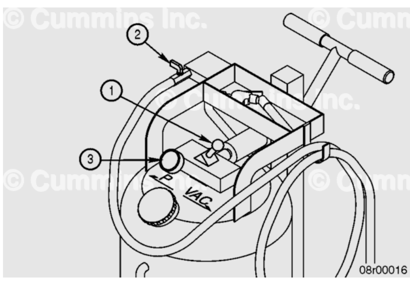
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



The coolant replacement tool tank capacity of 68 liters [18 gal] is adequate for most applications. An additional storage tank can be used for cooling system packages with more than 68 liter [18 gal] capacity.

Be sure there is no air pressure in the coolant replacement tool tank by opening the pressure relief valve (1) located on the control block of the coolant replacement tool.



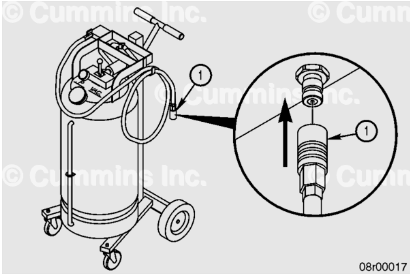
Connect the coolant replacement tool to a shop air supply regulated at 621 kPa [90 psi].

Switch the coolant replacement tool control lever (1) to “VAC” and leave the service hose valve (2) closed. This will create a vacuum in the tank to evacuate coolant from the cooling system package.

Monitor the gauge (3) and build a vacuum of approximately 508 mm-Hg [20 in-Hg] in the coolant replacement tool tank. Once the vacuum has been achieved, move the control valve lever (1) to the middle position.

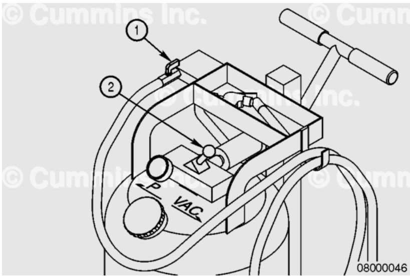
Maintain approximately a 508 mm-Hg [20 in-Hg] vacuum to achieve a faster drain.

Attach the fill hose quick disconnect coupling (1) of the coolant replacement tool to the quick disconnect fitting. Location of the fitting may vary between original equipment manufacturers (OEMs), but the fitting is generally located in the lowest point of the vehicle/equipment cooling system package.



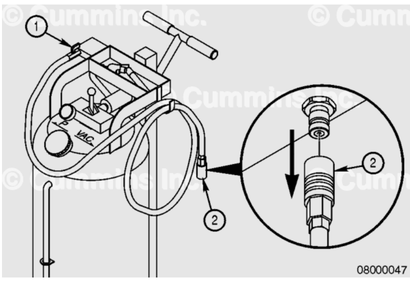
Note : Most Volvos are equipped with a different style fitting located in the radiator. An adapter hose is needed to connect the Cummins® coolant replacement tool to the fitting. The adapter hose is included in the accessory kit.

Open the service hose valve (1) by turning it **clockwise** until it is completely open (approximately ¼ turn).



Additional shop air may be required to maintain enough of a system vacuum to remove the coolant from the system. This can be done by moving the control valve lever (2) back to the VAC position.

When the cooling system has been evacuated, a coolant and air mixture will be visible in the clear section of the coolant replacement tool fill hose.



Note : Some residual coolant will settle in the coolant package as the recessed areas of the block continue to drain down over the next few minutes.

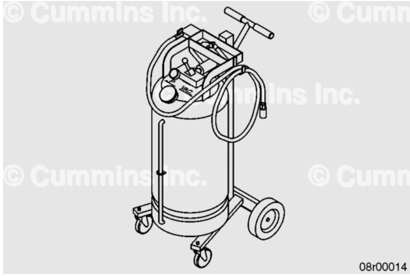
Once the system has been evacuated, turn the service hose valve (1) to the closed position by turning the valve **counterclockwise** a ¼ turn. Disconnect the shop air connection from the coolant replacer tool.

Disconnect the fill hose quick disconnect coupling (2) on the coolant replacement tool from the quick disconnect fitting.

Injection

⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.



The following steps are used to inject coolant into the cooling system using the coolant replacer tool, Part Number 2892459.

Note : When the vehicle/equipment or engine is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method is the preferred method for coolant removal. Use the coolant replacer tool, Part Number 2892459. If the vehicle/equipment or engine does not have a quick disconnect fitting presently installed, one can be installed in the cooling system package to utilize the Coolant Replacer Method; otherwise the coolant drain and fill method must be used.

A premixed, heavy duty coolant, or a mixture of high quality water and heavy duty coolant, can be used if the coolant used meets Cummins® Engineering Standards (CES). Coolant must meet the appropriate CES. Refer to Procedure 379-004 in Section 5 (/qs3/pubsys2/xml/en/procedures/00/00-379-004.html) of the Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Good quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as ($CaCO_3$ + $MgCO_3$)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO_4)

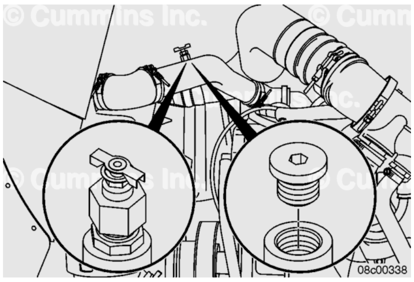
Cummins Inc. recommends the use of Fleetguard® products.

See the Fluids for Cummins® Products Service Manual, Bulletin 5411406, for more engine coolant specifications.

⚠ CAUTION ⚠

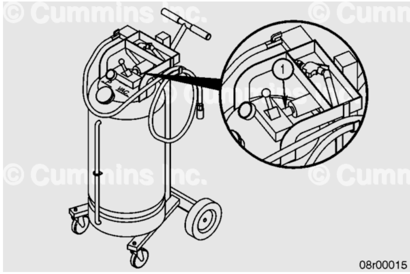
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.

Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

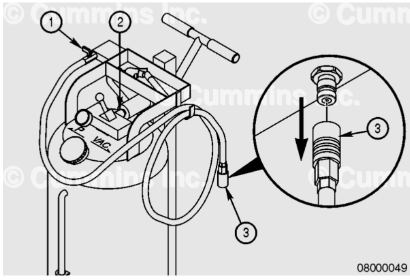


If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

Be sure there is no air pressure in the coolant replacement tool tank by opening the pressure relief valve (1) located on the control block of the coolant replacement tool.



Connect the coolant replacement tool to a shop air supply regulated at 621 kPa [90 psi].



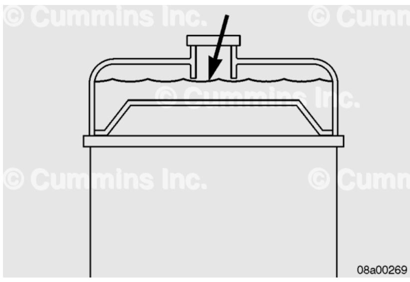
Attach the fill hose quick disconnect coupling (1) of the coolant replacement tool to the quick disconnect fitting located in the vehicle/equipment cooling system package.

With the service hose valve (2) in the closed position, switch the coolant replacement tool control lever (3) to “P” and build 172 kPa [25 psi] pressure on the gauge.

Slowly open the service hose valve (2) part way by turning it **counterclockwise** approximately 1/8 turn.

Do **not** open the ball valve completely as this will increase the coolant flow rate and increase the amount of air pockets created in the cooling system. This will provide a more complete injection of coolant.

Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill or recovery/expansion tank or until the coolant replacer tool is empty (whichever occurs first).



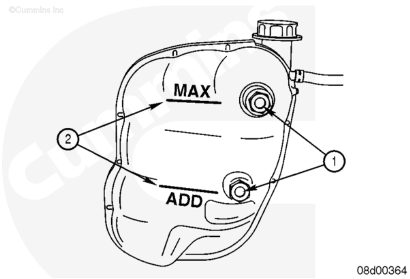
On applications that use a coolant recovery system, check to make sure the coolant is at the appropriate level in the coolant recovery tank for the engine temperature.

Many coolant recovery/expansion tanks, also called "top tanks", have sight glasses or are made of a clear material (**not** shown) to aid in checking the coolant level without removing the radiator cap.

It is important to understand the impact of temperature on the expansion of the coolant. Most "top tanks" do **not** have a provision for a "FULL HOT" coolant level. Filling the "top tank" while hot will result in a low operating level once the system has cooled.

⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.



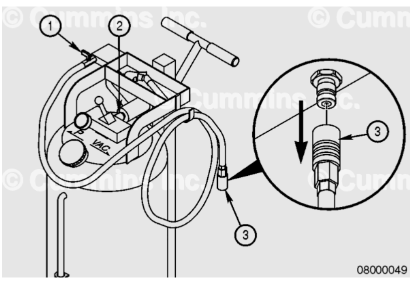
Note : If all coolant drained from the system was collected, the same volume or more must go back into the system. If any drained coolant remains in the tool after filling, this is an indication of an air pocket in the cooling system package which must be purged before returning the vehicle to service.

Note : Top off of coolant might be necessary for repairs that were performed to correct a coolant loss issue.

If all coolant drained from the system would **not** return to the system or the level is above the maximum level. This is an indication of an air pocket in the cooling system package, which **must** be purged before returning the vehicle to service.

To remove an air pocket from the cooling system, the coolant replacement tool can be used. See the vacuum section of this procedure.

Once the coolant level has been returned to the correct level close the service hose valve (1) by turning the valve **clockwise** until closed.



Remove pressure from the coolant replacement tool tank by opening the pressure release valve on the back of the control block (2).

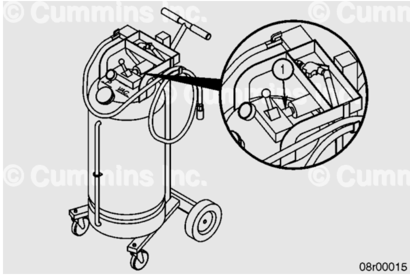
Disconnect the service hose quick disconnect coupling (3) from the quick disconnect fitting of the vehicle/equipment.

Vacuum

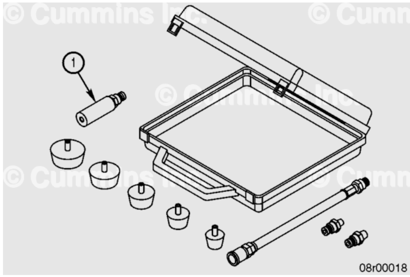
The following steps are used to place a vacuum on the cooling system using the coolant replacement tool, Part Number 2892459.

Be sure there is no air pressure in the coolant replacement tool tank by opening the pressure relief valve (1) located on the control block of the coolant replacement tool.

Clamp off any vent hoses/connections or overflow to the cooling system.



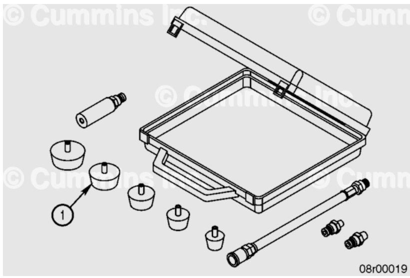
Attach the coolant dam handle (1) to the fill hose of the coolant replacement tool.



Attach the appropriate size coolant dam rubber adapter (1) onto the coolant dam handle.

The size of the fill neck will differ between OEMs choose the appropriate sized coolant dam rubber adapters.

Connect the coolant replacement tool to shop air regulated at 621 kPa [90 psi].



Place the coolant dam over the coolant fill neck in the radiator or overflow tank.

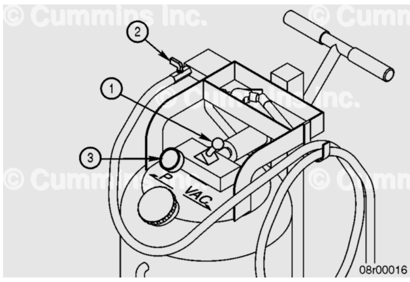
The size of the fill neck will differ between OEMs. Choose the appropriate sized coolant dam rubber adapters.

Switch the coolant replacement tool control lever (1) to “VAC” and leave the service hose valve (2) closed. This will create a vacuum in the tank to evacuate coolant from the cooling system package.

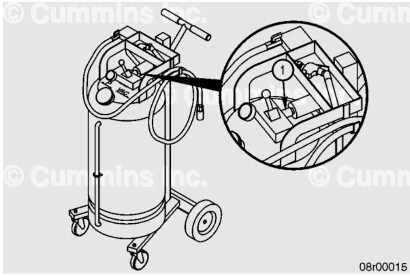
Monitor the gauge (3) and build a vacuum of approximately 508 mm-Hg [20 in-Hg] in the coolant replacement tool tank.

Slowly open the service hose valve (2) by turning it **clockwise** until it is completely open approximately ¼ turn.

Once the cooling system is put into a vacuum, any air trapped in the cooling system will be evacuated through the top of the system. This is noticed as air bubbles through the overflow tank or top of the radiator.



When air bubbles are no longer being drawn to the top of the cooling system move the control lever to the middle position. Remove the vacuum on the coolant replacement tool tank by opening the pressure relief valve located on the control block of the coolant replacement tool (1).



Remove the coolant dam from the radiator fill neck or overflow tank.

Disconnect the coolant replacement tool from the regulated shop air supply.

Install the radiator cap.

Drain

Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

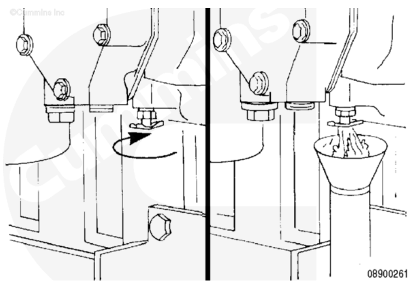
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Avoid excessive contact, and wash thoroughly after contact.
- Keep out of reach of children.

⚠ CAUTION ⚠

Use caution when draining coolant that coolant is not spilled or drained into the bilge area. Do not pump the coolant overboard. If the coolant is not reused, it must be discarded in accordance with local environmental regulations.

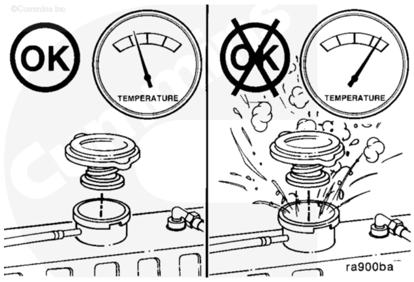


Drain the engine cooling system by opening the drain valve on the engine oil cooler (exhaust side of engine). A drain pan with a capacity of 31.7 liters (8 gal) will be adequate in most installations. Remove the pressure cap to allow the coolant to drain properly. After the cooling system is completely drained, close the drain valve.

All Applications Except Marine

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

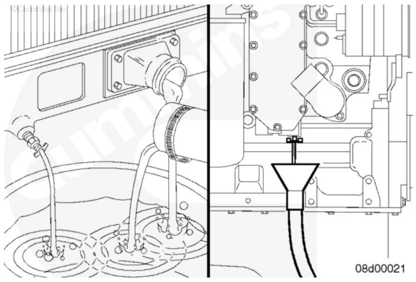


⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Avoid excessive contact, and wash thoroughly after contact.
- Keep out of reach of children.

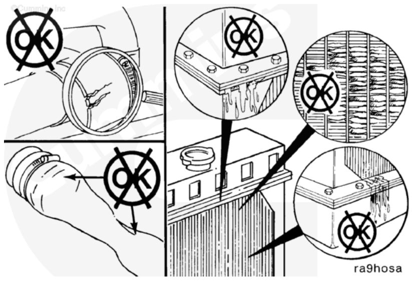
Drain the cooling system by opening the drain valve on the radiator and removing the plug in the bottom of the water inlet hose. A drain pan with a capacity of 19 liters [5 gal] will be adequate for most applications.



Check for damaged hoses and loose or damaged hose clamps. Replace as required.

Check the radiator for leaks, damage, and buildup of dirt.

Clean and replace as required.



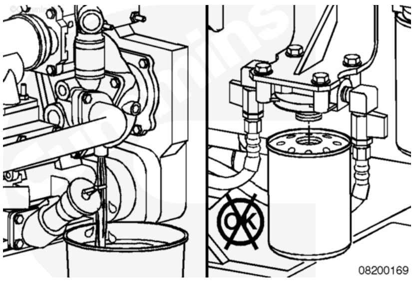
Flush

Marine Applications

RESTORE™ is a heavy-duty cooling system cleaner that removes corrosion products, silica gel, and other deposits. The performance of RESTORE™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times or the use of RESTORE Plus™. Up to twice the recommended concentration levels of RESTORE™ can be used safely. RESTORE Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

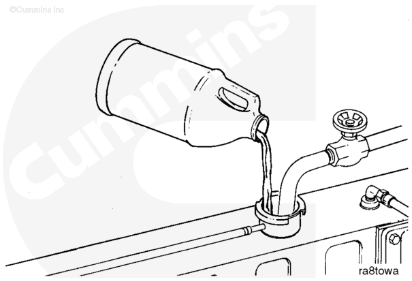


If **not** previously done, drain the cooling system. See the information above in "Cooling System - Drain” step. Do **not** allow the cooling system to dry out.

Do **not** remove the coolant filter.

⚠ CAUTION ⚠

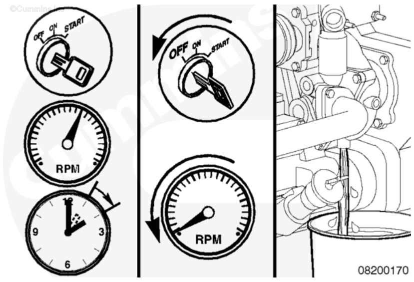
Fleetguard® RESTORE™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operations.



Immediately add 3.8 liters [1 gal] of Fleetguard® RESTORE™, RESTORE Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity. Fill the system with clean water.

Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

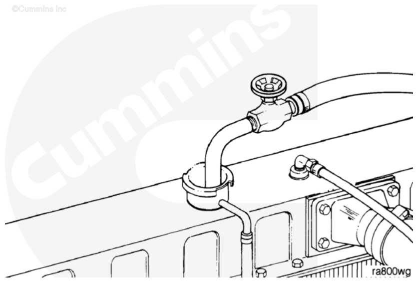
Shut the engine OFF, allow to cool to 50°C [122°F], and drain the cooling system.



The system has a design fill rate of 19 liters per minute [5 gallon per minute].

Marine engines **must** be vented at the coolant outlet housing and at the rear of the expansion tank during filling.

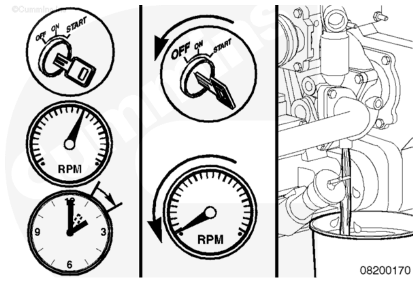
Fill the cooling system with clean, good quality water.



Operate the engine for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF, allow to cool to 50°C [122°F], and drain the cooling system.

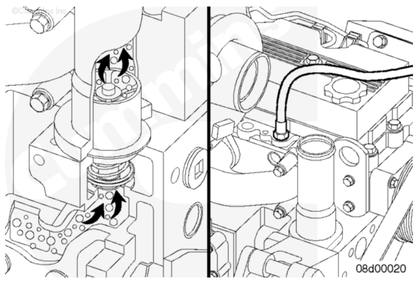
If the water being drained is still dirty, the system **must** be flushed again until the water comes out clean.



All Applications Except Marine

⚠ CAUTION ⚠

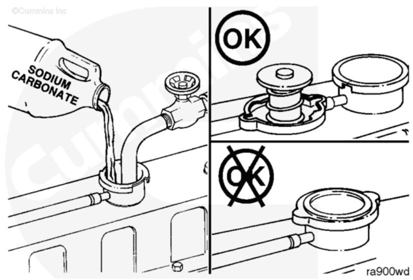
The system must be filled properly to prevent air locks. During filling, air must be purged from the engine coolant passages. Be sure to open the petcock on the aftercooler for aftercooled engines. Wait 2 to 3 minutes to allow the air to be vented; then add coolant mixture to bring the level to the top.



Note : Adequate venting is provided for a fill rate of 19 liters [5 gal] per minute.

⚠ CAUTION ⚠

Do not install the radiator cap. The engine is to be operated without the cap for this process.

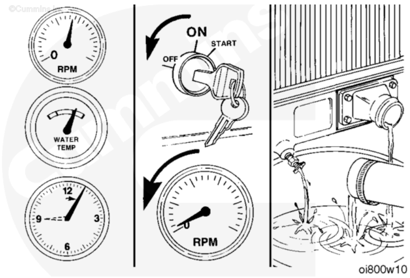


Fill the system with a mixture of sodium carbonate and water (or a commercially available equivalent).

Note : Use 0.5 kg [1 lb] of sodium carbonate for every 23 liters [6 gal] of water.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



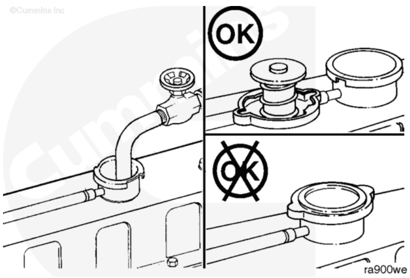
Operate the engine for 5 minutes with the coolant temperature above 80°C [176°F].

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

Note : Make sure to vent the engine and aftercooler for complete filling.

Note : Do not install the radiator cap or the new coolant filter.

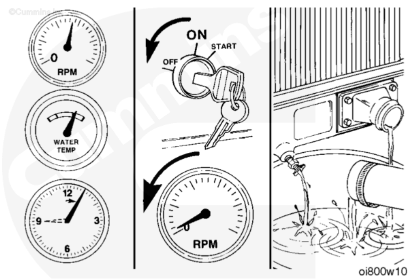
Fill the cooling system with clean, good, and quality water.



Operate the engine for 5 minutes with the coolant temperature above 80°C [176°F].

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

Note : If the water being drained is still dirty, the system must be flushed again until the water comes out clean.



Fill

Marine Applications

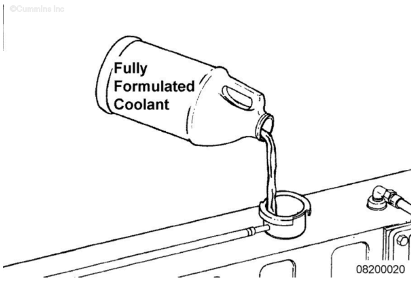
QSC8.3 and QSL9 Engines with Heat Exchangers

The system has a design fill rate of 19 liters per minute [5 gallons per minute].

Marine engines **must** be vented at the coolant outlet housing, and at the rear of the expansion tank during filling.

Fill the cooling system with fully formulated coolant or a 50/50 mixture of recommended antifreeze and clean, good quality water, as outlined in the Owners Manual.

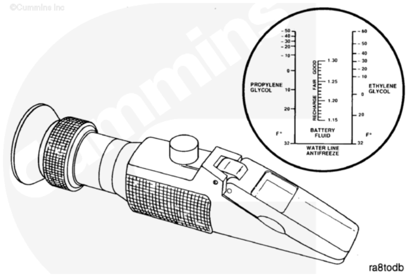
Install the pressure cap. Operate the engine to 50°C [122°F], and check for coolant leaks.



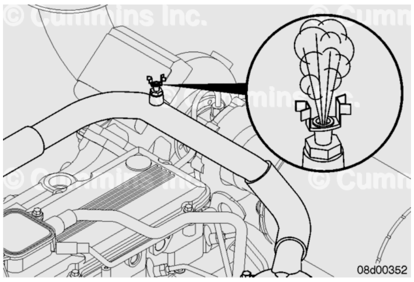
The freeze point protection **must** be checked after coolant is added to the cooling system.

The Fleetguard® refractometer, Part Number C2800, provides a reliable, easy-to-read, and accurate measurement of freeze point protection and antifreeze concentration.

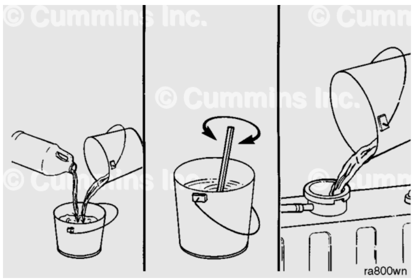
See equipment manufacturer service information for the correct operation instructions for the Fleetguard® refractometer, Part Number C2800.



On QSL9 Keel Cooled Engines, open all engine mounted petcocks and remote heater petcocks.



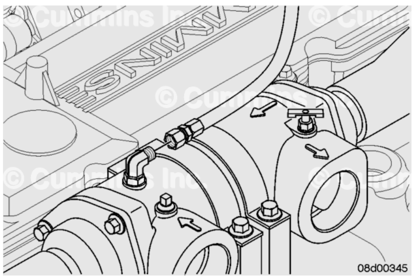
Fill the cooling system with fully formulated coolant or a 50/50 mixture of recommended antifreeze and clean, good quality water, as outlined in the Owners Manual.



Note : The timing and sequence of this action will depend on the height of the petcocks in the cooling system.

Fill the engine until a solid stream of coolant is visible from the keel cooler thermostat housing, heat transfer tube, and remote mounted heater petcocks. Close the petcocks.

Fill until the coolant expansion tank is filled to the proper level and the low coolant level sensor is covered.



Start the engine. Monitor the coolant level and the low coolant alarm until all air is purged from the system. Operate the engine for 5 minutes with the pressure cap removed, however, do **not** exceed 50°C [122°F].

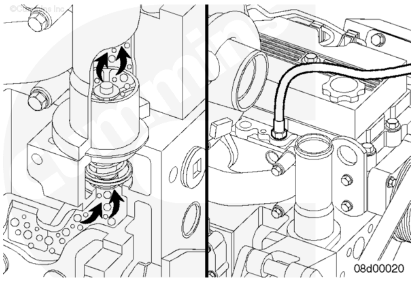
Shut down the engine and check the coolant level again. Add coolant if needed.

Install the pressure cap. Start the engine and monitor the low coolant alarm during the initial startup test.

All Applications Except Marine

⚠ CAUTION ⚠

The system must be filled properly to prevent air locks. During filling, air must be purged from the engine coolant passages. Be sure to open the petcock on the aftercooler for aftercooled engines. Wait 2 to 3 minutes to allow the air to be vented; then add coolant mixture to bring the level to the top.



⚠ CAUTION ⚠

During all coolant fill procedures, all coolant flow valves to equipment heating systems must be opened in order to purge air from those systems as well as from the base engine cooling system. These valves must remain open during the engine cooling system de-aeration process. Make sure adequate coolant levels are maintained in the coolant reservoir during the entire fill procedure. Special care must be taken when filling EGR cooler-equipped engines to make sure all air is purged from the system.

The system is designed to use a specific quantity of coolant. If the coolant level is low, the engine will run hot.

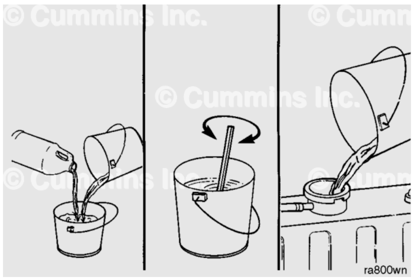
If frequent addition of coolant is necessary, the engine or system has a leak. Find and repair the leak.

The system has a designed fill rate of 19 liters [5 gal] per minute.

⚠ CAUTION ⚠

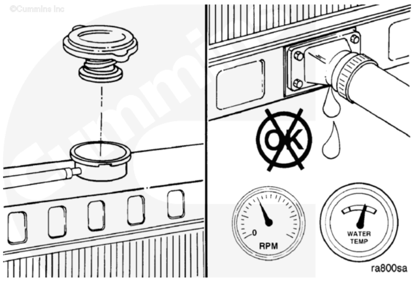
Never use water alone for coolant. This can result in damage from corrosion.

Use a mixture of 50 percent water and 50 percent ethylene glycol or propylene glycol antifreeze to fill the cooling system.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Install the pressure cap. Operate the engine until the coolant reaches a temperature of 80°C [176°F]. Check for coolant leaks.

Operate the engine at idle for 25 minutes to purge the air from the cooling system.

Turn all cab heater temperature switches, if equipped, to high to allow maximum coolant flow through the heater core(s). The blower does **not** have to be on.

Check the coolant level again to make sure the system is full of coolant or that the coolant level has risen to the hot level in the recovery bottle on the system, if so equipped.

Last Modified: 15-Jan-2019